

Azure Architecture

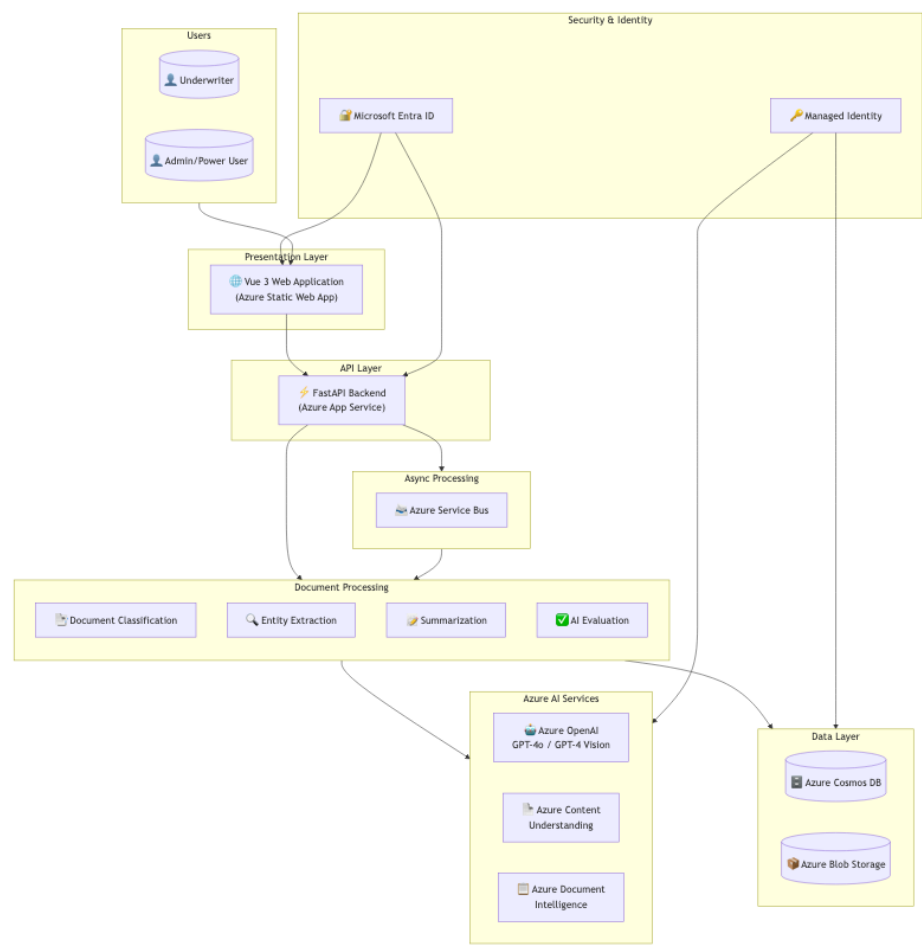
ISE Team

HNW Insurance Intelligence Platform - Azure Architecture

Overview

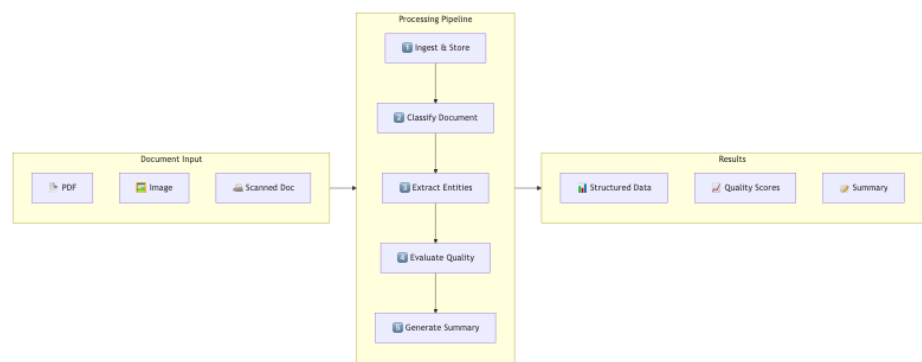
This document describes the Azure architecture for the Document Intelligence Platform, designed for insurance underwriting automation with extensibility for other document processing use cases.

High-Level Architecture Diagram



diagram

Component Architecture



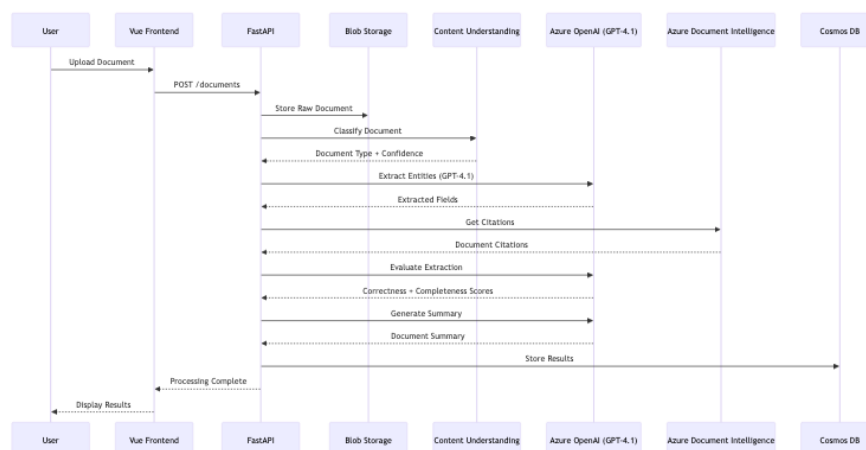
diagram

Azure Services Used

Service	Purpose
Azure App Service	Host FastAPI backend

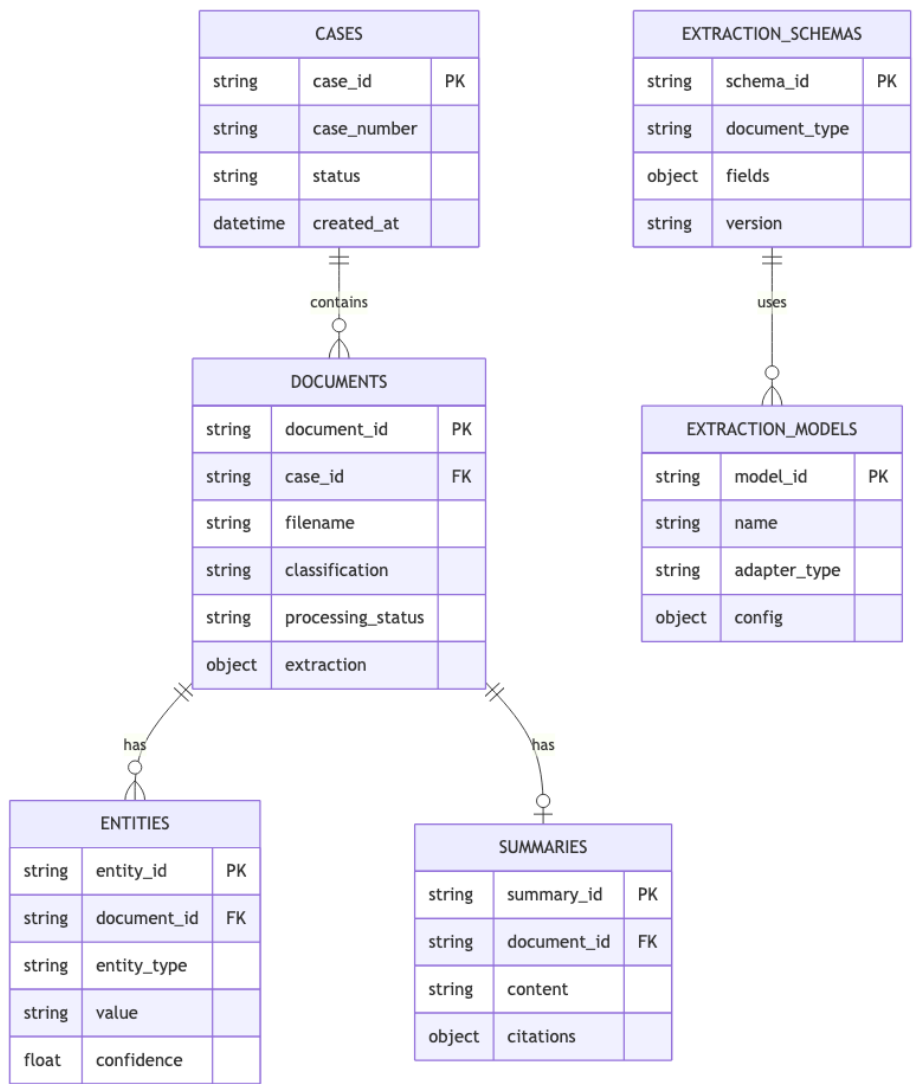
Service	Purpose
Azure Static Web Apps	Host Vue 3 frontend
Azure Cosmos DB	Store cases, documents, entities, schemas
Azure Blob Storage	Store raw documents and processed results
Azure OpenAI Service	GPT-4.1 for extraction
Azure Content Understanding	Document classification
Azure Document Intelligence	OCR, layout extraction, and citations
Azure Service Bus	Async message processing
Microsoft Entra ID	Authentication and authorization
Azure Key Vault	Secrets management
Azure Monitor	Logging and observability

Data Flow



diagram

Cosmos DB Collections



diagram

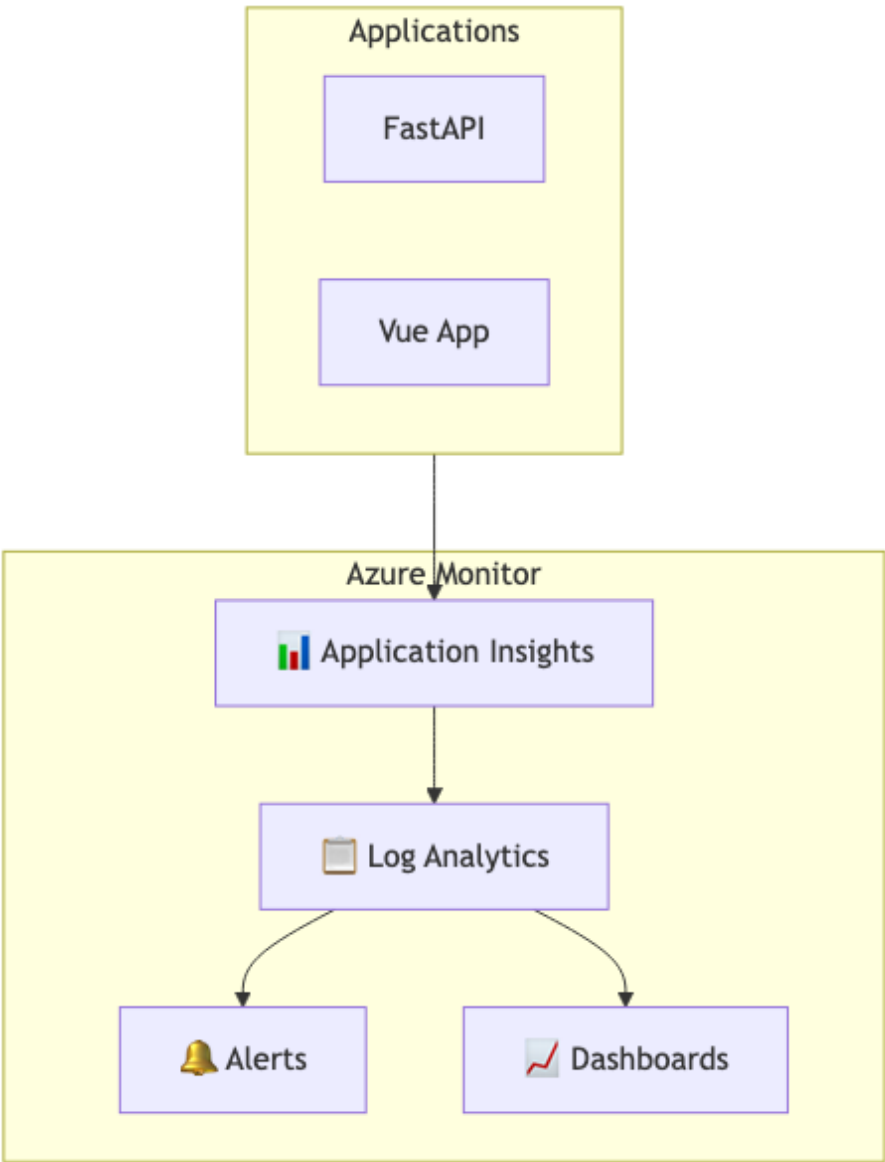
Security Architecture

Layer	Security Control
Identity	Microsoft Entra ID (OAuth 2.0 / OIDC)
API Access	Managed Identity (no stored secrets)
Network	Private endpoints, NSGs, WAF
Data at Rest	Azure-managed encryption (AES-256)
Data in Transit	TLS 1.3
Secrets	Azure Key Vault
Audit	Azure Monitor, Activity Logs

Scalability Considerations

Component	Scaling Strategy
App Service	Horizontal auto-scale based on CPU/Memory
Cosmos DB	Serverless with auto-scale RU/s
Blob Storage	Unlimited scale, geo-redundant option
Azure OpenAI	Rate limiting, request queuing
Service Bus	Partitioned queues for throughput

Monitoring & Observability



diagram

Architecture Version: 1.0
Last Updated: January 2026